# VARDHAMAN COLLEGE OF ENGINEERING, HYDERABAD

Autonomous institute affiliated to JNTUH

## DEPARTMENT OF CSE(AI&ML)

## IV-Year Major Project Abstract Level Review

**Title:** Adaptive Multi-Granularity Hierarchical Indexing for Retrieval in Heterogeneous Documents

**Domain:** Artificial Intelligence & Machine Learning (AI&ML) / NLP

BATCH ID : 48

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## Supervisor

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**Department of CSE (AI&ML)**

# Outlines

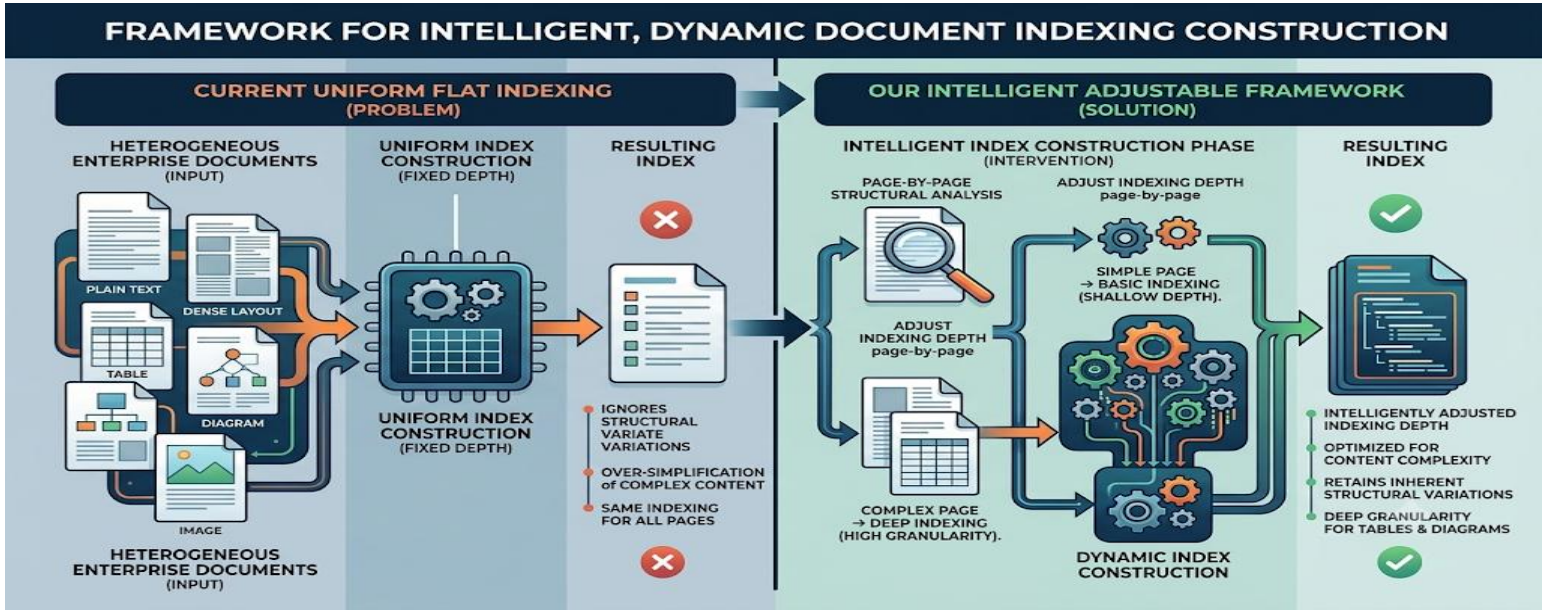
- Abstract
- Introduction
- Literature Review
- Existing System with Pros and Cons (comparison with existing systems- Table)
- Objectives & Expected Deliverables
- SDG Mapping
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- PSO Mapping
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# Abstract


- Traditional RAG systems treat all document pages identically, using fixed-size chunking.
- Proposes an **Adaptive Multi-Granularity Hierarchical Indexing** framework for heterogeneous documents.
- Evaluates page complexity using token density, layout, images, and tables.
- Dynamically assigns indexing depth: coarse nodes for simple pages, fine-grained subtrees for complex ones.
- A smart retrieval mechanism extracts precise context, preserving structural meaning.
- Reduces API token consumption while ensuring highly accurate retrieval for law, medicine, and finance.

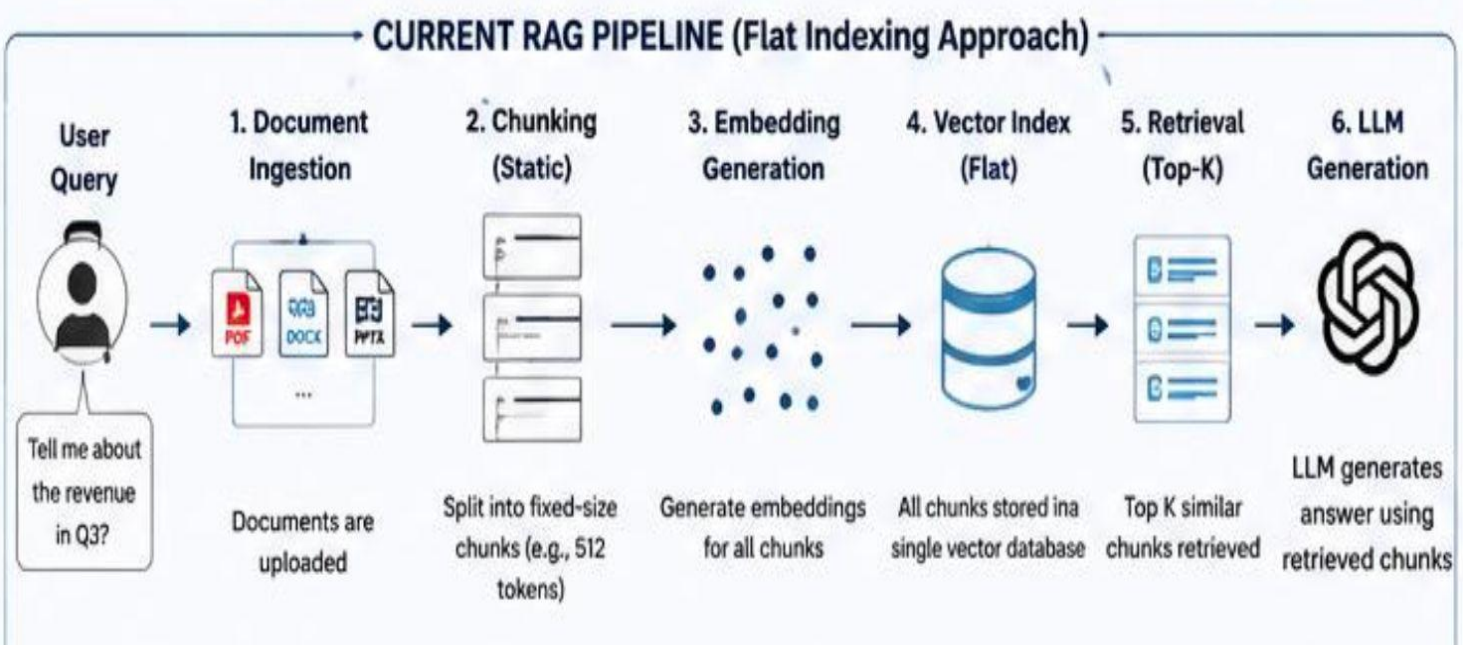
# Introduction

- Real-world enterprise documents are rarely uniform; they freely mix plain text with dense layouts (tables, diagrams, images).
- Current AI systems process these documents utilizing uniform, flat indexing.
- This approach ignores the inherent structural variations in document complexity.
- Our framework intervenes at the **index construction phase** (rather than just query routing) to intelligently adjust indexing depth page-by-page.



# Existing System

- Current models rely on a uniform, **flat indexing approach**.
- Documents are broken down into fixed-size chunks (e.g., 512 tokens).
- Treats a simple text page exactly the same as a dense page packed with complex tables or diagrams.
- Routing and decision-making usually occur at the query stage rather than during index construction.



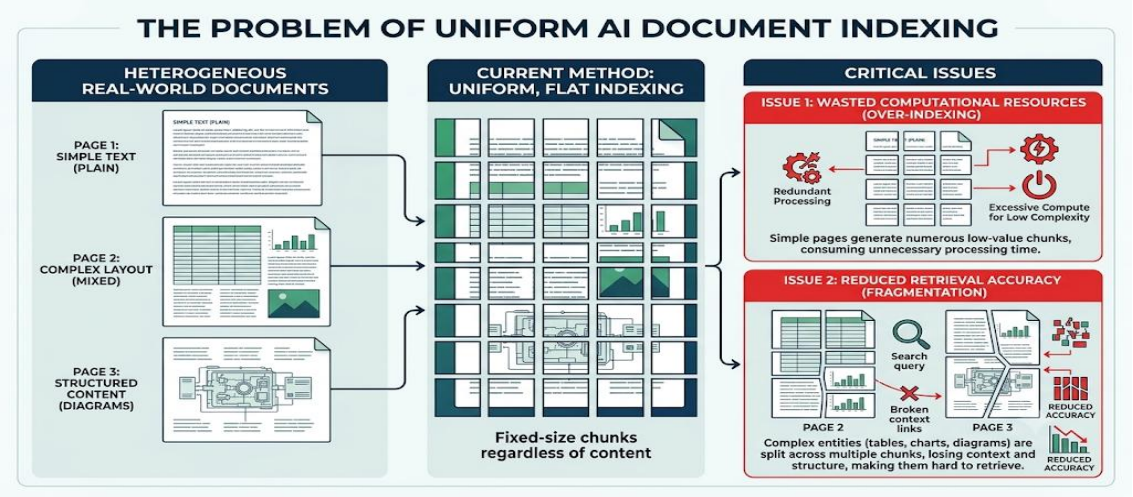
# Pros and Cons of Existing System

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- **Pros:**
  - Simple and easy to implement.
  - Works efficiently for highly homogeneous, plain-text documents.
- **Cons:**
  - **Resource Waste:** Assigns unnecessarily deep indexing to simple pages, wasting computational resources.
  - **Accuracy Loss:** Dense, complex pages are structurally fragmented and underrepresented.
  - **High Costs:** Fetches irrelevant contexts, increasing API token consumption.

# Problem Statement

- Real-world documents are highly heterogeneous, mixing plain text with dense layouts containing tables, diagrams, and images.
- Currently, AI systems process these documents using a uniform, flat indexing approach that breaks every page into fixed-size chunks regardless of structural complexity.
- This rigid methodology introduces two critical issues: it wastes computational resources by over-indexing simple pages, and it significantly reduces retrieval accuracy because complex pages are underrepresented and structurally fragmented.



# Objectives & Deliverables Proposed System

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- **Objectives:**
- **Obj 1:** Develop a complexity-scoring module based on token density, tables, images, and layout intricacy.
- **Obj 2:** Implement an adaptive tree-building algorithm for hierarchical vector indexing (coarse vs. fine-grained).
- **Obj 3:** Build a smart retrieval system to navigate the multi-granularity hierarchy to fetch precise contexts.
- **Expected Deliverables:**
- An adaptive document ingestion and indexing module.
- A functional RAG prototype interface that improves retrieval accuracy and reduces API token usage.

# SDG Mapping

- **SDG 4 (Quality Education):** Accurately answers student queries from complex textbooks at the correct detail level.
- **SDG 9 (Industry, Innovation & Infrastructure):** Provides highly scalable document retrieval for dense engineering and financial documentation.
- **SDG 12 (Responsible Consumption & Production):** Reduces computational waste and AI energy footprint by avoiding over-indexing.
- **SDG 17 (Partnerships for the Goals):** Enables seamless information retrieval across diverse global research, medical, and legal teams.












# PO Mapping

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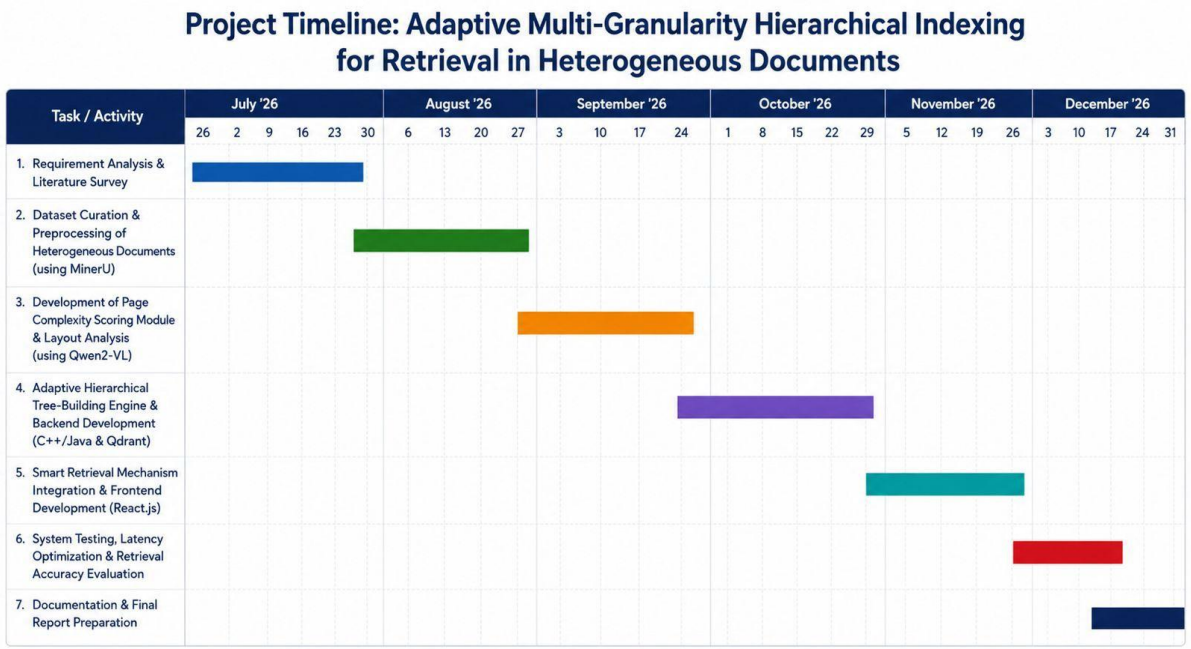
- **PO1 (Engineering Knowledge):** Applying algorithms to build adaptive hierarchical trees.
- **PO2 (Problem Analysis):** Analyzing inefficiencies of fixed-size chunking.
- **PO3 (Design/Development):** Tailoring representation depth based on per-page complexity.
- **PO5 (Tool Usage):** Utilizing modern NLP/Vision models and vector databases.
- **PO7 (Environment):** Minimizing unnecessary computational token usage.
- **PO8 (Ethics):** Ensuring factual fidelity in high-stakes fields.
- **PO11 (Project Management):** Cutting API costs for large-scale enterprise discovery.

# PSO Mapping

- **PSO1:** Applying AI solutions to resolve complex queries in diverse domains like legal contracts, clinical records, and financial reports.
- **PSO2:** Demonstrating expertise in using advanced ML tools, specifically vision-language models and layout analysis to recognize page complexity and make intelligent routing decisions.

# Timelines

- **July-Aug '26:** Requirement Analysis & Dataset Curation (MinerU).
- **Sept-Oct '26:** Page Complexity Scoring (Qwen2-VL) & Adaptive Tree-Building Engine Backend.
- **Nov '26:** Smart Retrieval Mechanism & Frontend (React.js).
- **Dec '26:** System Testing, Accuracy Evaluation & Final Deployment.



# References


- R. Li et al., "SeerRAG: Seer-Like Routing for Uncertainty-Guided Retrieval-Augmented Generation," *IEEE*, 2025.
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# Thank you

